COOLSHADE · PRODUCT REQUIREMENTS AND SPECIFICATIONS

CoolShade product requirements and specifications

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/coolshade		
Source	docs/03-requirements.md @ ae091ec		

REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-30	Amish Chadha	Kit 1.9.1 upgrade: requirements renumbered PR-001 to PR-017 (the Was column keeps R1 to R17), each traced to a user requirement in CSH-URS-001; document is now the PRS (was CSH-REQ-001); R8 status updated for the white enclosure (CSH-DEC-001 D9); no target changed
0.7	2026-10-02	Amish Chadha	R14 tool list adds a hand rivet nut tool and a bench vice, decided by Amish on 2026-10-02
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-09-30	Amish Chadha	Status from CSH-CAL-001 v0.3 for the constructable design (CSH-DDR-003); R13 not met against $775, new budget proposed
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	

			TRL 3 status from CSH-CAL-001 for every requirement; R6 thresholds adopted under CSH-DDR-001; R13 stated against $700 and the proposed $750
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold, as CSH-REQ-001 (requirements R1 to R17)

CoolShade product requirements and specifications

What the canopy must do, each requirement traced to a user need in CSH-URS-001 (links in CSH-RTM-001). The requirements describe option A, the solar misting shade canopy, which is the current design; the concept options and the proposed choice are in CSH-OPT-001. The document was CSH-REQ-001 (requirements R1 to R17) before the kit 1.9.1 upgrade of 2026-10-30; the numbers are the same (PR-001 is R1) and the Was column keeps the old number. Targets are proposals for review, not yet validated with users, and will be revised after co-design sessions (CSH-PRB-001). Status comes from the TRL 3 calculations in CSH-CAL-001 v0.6 for the constructable design of CSH-DDR-003: **13 of the 17 are met on paper or by design; PR-013 (cost) is over its value-engineering target (USD 882 against USD 775), PR-009 (water hygiene) and PR-016 (maintenance) are at risk, and PR-005 (no wetting) cannot be judged at TRL 3 for lack of droplet data.** Four of the 13 are met by design only (PR-006, PR-011, PR-012, PR-014) and PR-010 is met on paper.

Design day: 38 °C air, 25 % relative humidity, clear sky, wind about 1 m/s, 5 h of misting demand, 4.5 peak sun hours. **Humid edge:** 32 °C and 60 % relative humidity, the wettest air in which the controller still mists.

Table 1. Product requirements and specifications. Status after the TRL 3 calculations (CSH-CAL-001 v0.6, python docs/04-calcs/sizing.py; tags in brackets are lines of its output).

ID	Traces to	Category	Title	Requirement	Verification	Status	Was
PR-001	UR-001, UR-004	Performance	Shaded waiting area	6 m² or more of shade at solar noon; room for at least 6 people standing or seated and one wheelchair space of 0.8 x 1.2 m	Plan layout; sun path check per site	Met: shadow 6.30 m² or more at any sun elevation of 45° or more; room for 8 standing, 3 seated and a wheelchair [A2, A3]. Where the shadow falls depends on the site	R1
PR-002	UR-004	Physical architecture	Headroom	Clear height 2.2 m or more under any part of the frame, nozzle or brace in the waiting area	Model check	Met: 2,223 mm under the flattened ends of the rear knee braces; nozzle tips 2,559 mm or higher [B2, B3]	R2
PR-003	UR-001	Performance	Radiant heat reduction	Mean radiant temperature under the canopy at least 15 °C below full sun at midday on the design day	Literature comparison; later field measurement with a globe thermometer (HeatMap Node method)	Met on published evidence: about 17.3 °C for shade sails in Tempe (Middel et al., 2021); unverified for this fabric	R3
PR-004	UR-002	Performance	Air cooling from misting	Air temperature in the waiting zone (0.5 to 2.0 m height) at least 2 °C below ambient while spraying on the design day (restated, CSH-DDR-002; was averaged over the spray cycle)	Energy balance at TRL 3; later field test	Met: 2.71 °C while spraying at 1 m/s, held to 1.35 m/s; 1.35 °C averaged over the cycle [E2, E3]	R4
PR-005	UR-003	Performance	No wetting	No perceptible wetting of skin or clothing at 1.5 m height on the design day; no puddles on the waiting area	Droplet size data from nozzle maker; later field test	Not verifiable at TRL 3: met only if the nozzle's Dv0.9 is below about 80 µm at 7 bar; no droplet data yet [D3]	R5
PR-006	UR-005, UR-003	Performance	Smart activation	Mists only when air temperature is 32 °C or more, relative humidity is 60 % or less, presence is detected in the last 2	Control logic review; later bench test	Met by design: stops within 30 s [I1]; thresholds decided (CSH-DDR-001, D2)	R6

ID	Traces to	Category	Title	Requirement	Verification	Status	Was
				min and the tank is above its low level; stops within 60 s when any condition fails			
PR-007	UR-015	Performance	Water use	100 L or less per design day	Nozzle flow calculation	Met: 85 L used, 95 L drawn with the drain-and-refill routine [C2, C5]; holds up to 4.75 L/h per nozzle [C3]	R7
PR-008	UR-006	Performance	Energy autonomy	Energy-neutral on the design day from the onboard panel; battery covers one full design day with no sun	Energy budget	Met: 203.8 Wh used against 337.5 Wh harvested; 25 Ah battery (CSH-DDR-002; was 20 Ah, 1.005 days) covers 1.256 design days; with the white enclosure (CSH-DEC-001 D9) no charging in air above about 42.7 °C [F2, F5]	R8
PR-009	UR-007	Safety	Water hygiene	Potable-quality supply water; 5 µm filtration; mist line drained within 5 min after every session; water in the tank 48 h old or less; opaque, shaded, lockable tank; documented weekly disinfection	Design review; later microbiological sampling with a partner lab	**At risk:** line drains in about 30 s with the vacuum breaker and water stays under 24 h old with the routine [C5, C7], but tank water sits at about 34 °C, in the 20 to 50 °C *Legionella* growth range (WHO) [I3]	R9
PR-010	UR-008	Safety	Wind resistance	Frame and anchors survive a 3 s gust of 30 m/s (108 km/h) with the fabric removed; fabric fitted only when forecast gusts are below 15 m/s (54 km/h); fabric removable in 15 min or less (restated, CSH-DDR-002; was 30 m/s with the fabric fitted)	Structural calculation at TRL 3	Met on paper: fabric off at 30 m/s, posts 0.14, panel bearer 0.02 and purlins 0.07 utilization [G13]; fabric on at 15 m/s, long beams 0.88 [G14]; anchors need about 2.8 kN each, to be confirmed per site.	R10

ID	Traces to	Category	Title	Requirement	Verification	Status	Was
						The original case gives 3.52 on the beams [G7]	
PR-011	UR-010	Safety	Privacy	No camera or microphone; presence by passive infrared only; if a status link is fitted, only counts and levels (on-time, water used, temperature, humidity, faults) leave the device	Design review	Met by design; no status link fitted (CSH-DDR-001, D7)	R11
PR-012	UR-009, UR-006	Electrical	Electrical safety	12 V DC only, no mains; battery fused at the terminal; all electronics in a lockable IP65 enclosure; LiFePO4 chemistry with BMS and cold-charge cutoff	Design review	Met by design: 15 A terminal fuse, margin 2.0 on the largest current [I2]	R12
PR-013	UR-014	Cost	Cost	Estimated parts cost for one prototype within the $775 value-engineering target (`budget_usd` , a hypothetical control target set under CSH-DDR-002; was $700)	BOM	**Over the value-engineering target by $107:** estimated $882, 13.8 % over, after the parts added to make the design buildable (CSH-DDR-003); savings worth trying are in CSH-DEC-001 [J1, J2]	R13
PR-014	UR-012	Physical architecture	Buildability	Bolted assembly; no welding; built with a drill, angle grinder, spanners, a hand rivet nut tool, a bench vice and a ladder by two people in 2 days or less	Assembly review	Met by design: 79 bolts and anchors, 36 of them into rivet nuts, no welds [H2]; the hand rivet nut tool and bench vice were added to the tool list by Amish on 2026-10-02 (CSH-DEC-001); build time unverified	R14
PR-015	UR-011, UR-015	Usability	Refill interval	1 design day or more between tank refills	Water balance	Met: 1.41 design days per full tank [C4]	R15

ID	Traces to	Category	Title	Requirement	Verification	Status	Was
PR-016	UR-011	Usability	Maintenance	Nozzles cleanable or replaceable without tools in 15 min or less; filter changed in 10 min or less; descaling interval 1 month or longer	Design review; later field log	**At risk** in hard-water areas: scale can block 0.4 mm orifices; not verifiable at TRL 3	R16
PR-017	UR-013	Packaging and transport	Movability	Can be unbolted and moved to another site by two people in 1 day or less; no part heavier than 40 kg to lift (tank empty)	Mass estimate	Met: heaviest part a front post, 21.0 kg [H1]	R17

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software, cost. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

- Nozzle flow 4 L/h each at 7 bar, eight nozzles, 50 % spray duty cycle while active; this implies a discharge coefficient of 0.24 for a 0.4 mm orifice, to be checked against nozzle datasheets.
- About 87 % of the sprayed water evaporates in the air; the rest drifts away, drips or wets surfaces (estimate).
- Cooling is estimated over a control volume 3.0 m long and 2.0 m high with air crossing it at the wind speed. Real mixing is less even than this.
- Panel output 100 W x 4.5 peak sun hours x 0.75 system factor; battery 12.8 V 25 Ah, 80 % usable; enclosure white or light grey (solar absorptance 0.25, CSH-DEC-001 D9).
- Wind: fabric treated as solid with 5 % sag, net normal force coefficient 1.2, S275 steel with a 1.5 factor on wind. All other assumptions are listed in CSH-CAL-001, Table 1.